

- RAG Learning

6.1 Sampling, Charts & Scatter Graphs

AQA GCSE Mathematics 8300 — Statistics — spec refs S1, S2, S5, S6 — Papers 1 (non-calculator), 2 and 3 (calculator)

Everything in this topic is assessed on both tiers This topic is all Foundation and Higher content — there is no Higher-only material in S1, S2, S5 or S6; there is no Higher-only content in S1, S2, S5 or S6. It covers how data is collected from samples, how to display it in tables and charts, and how to analyse two variables with scatter graphs.

6.1.1 Populations and Samples (S1, S5)

Population — the whole group of people or items you want to find out about (e.g. all 1200 pupils in a school, every bolt made in a factory today).

Sample — a smaller part of the population that is actually surveyed, measured or tested.

A **census** surveys every member of the population.

Why use a sample?

- It is **quicker** and **cheaper** than surveying everyone.
- The population may be too large, or it may be impossible to list everyone.
- Testing may **destroy** the item (e.g. testing how long light bulbs last, crash-testing cars).

Limitations of sampling

- A sample gives only an **estimate** — different samples give slightly different results.
- A sample that is too **small** may not represent the population well.

- A sample may be **biased** — some groups are over- or under-represented.

Random sample — every member of the population has an **equal chance** of being selected.

How to take one: list and number every member of the population (this list is the *sampling frame*), then use a random number generator, or draw numbers from a hat, to choose the required number of members. Ignore repeated numbers.

Biased sample — a sample that does not fairly represent the population.

Common sources of bias

Sampling method	Why it is biased
Asking people at a bus station how often they use the bus	Bus users are over-represented
Asking your friends	Not random; friends tend to share views and age
Surveying only one street, one time of day, or one day of the week	Other areas/times are not represented
Postal or online questionnaires	Only people who choose to reply are included (non-response)
Taking the first 20 items off a production line	The machine may behave differently at different times

Exam Tip: When asked to criticise a sample, say *why* it is not representative in the context of the question (e.g. "people at a gym are likely to exercise more than the general population"), not just "it is biased". Improvements usually mean: **make it random**, **make it larger**, or **cover the whole population** (all areas, times, ages).

Using a sample to describe a population (S5)

If the sample is random and reasonably large, the proportion in the sample is a good estimate of the proportion in the population.

$$\text{Estimate for population} = \frac{\text{number in sample with property}}{\text{sample size}} \times \text{population size}$$

Worked Example: A school has 1200 pupils. In a random sample of 50 pupils, 12 walk to school. Estimate the number of pupils in the school who walk.

$$\text{Proportion in sample: } \frac{12}{50} = 0.24$$

$$\text{Estimate: } 0.24 \times 1200 = 288 \text{ pupils.}$$

Assumption: the sample is representative of the whole school.

Worked Example: A factory makes 8000 bulbs a day. A random sample of 250 bulbs contains 6 faulty bulbs.

$$\frac{6}{250} \times 8000 = 0.024 \times 8000 = 192 \text{ faulty bulbs per day (estimate).}$$

A bigger sample would give a more reliable estimate.

6.1.2 Tables and Frequency Tables (S2)

Categorical data is described by words or categories (e.g. favourite colour, type of pet).

Discrete numerical data can only take particular values, usually counted (e.g. number of siblings).

A **frequency table** records how many times each category or value occurs. A **tally** groups marks in fives (four strokes and a diagonal "gate").

Worked Example: The number of goals scored by a team in 20 matches is recorded.

Goals	Tally	Frequency
0		4
1		7
2		5
3		3
4		1
Total		20

Always check the frequencies add up to the total number of data values.

Two-way tables show two categories at once (e.g. gender and favourite subject); rows and columns must add to the totals. These are also used in probability (topic 5).

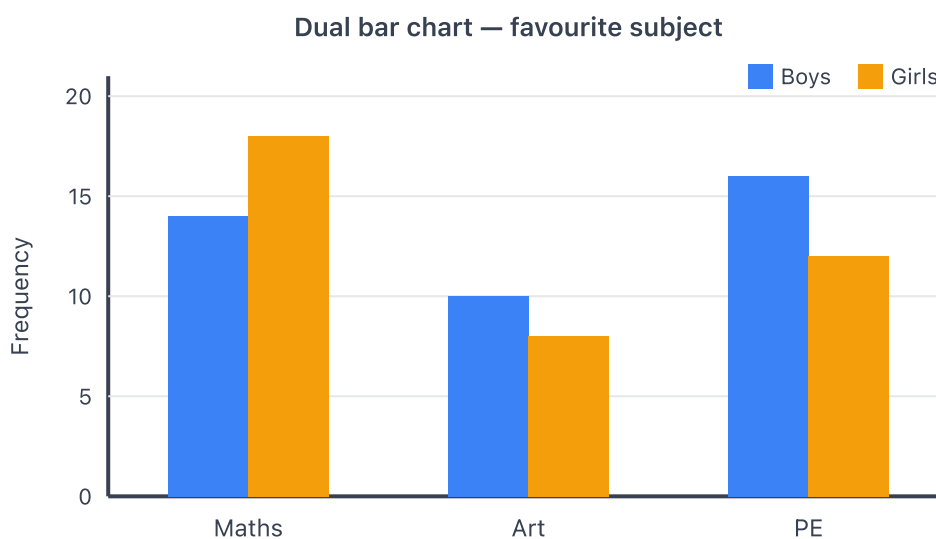
6.1.3 Bar Charts and Vertical Line Charts (S2)

Bar chart — bars of **equal width** with **equal gaps**; the height of each bar shows the frequency. Used for categorical or discrete data.

Dual (comparative) bar chart — two or more bars side by side for each category, with a key (e.g. boys and girls).

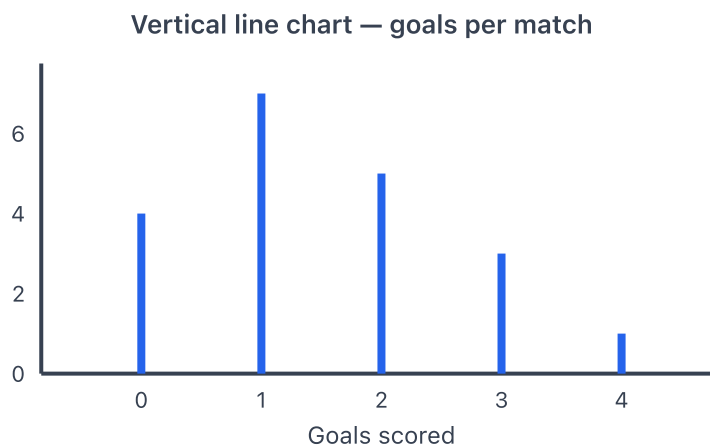
Composite (stacked) bar chart — each bar is split into parts that add to the total for that category.

Vertical line chart — like a bar chart but drawn with thin lines; used for **ungrouped discrete numerical data**.



Worked Example: From the dual bar chart: how many more students chose maths than art?

Maths: $14 + 18 = 32$. Art: $10 + 8 = 18$. Difference = $32 - 18 = 14$ students.



Exam Tip: Always label both axes, give the chart a title, use a sensible linear scale starting at 0, and keep bars the same width. In a bar chart for *categories* leave gaps between the bars.

6.1.4 Pictograms (S2)

A **pictogram** uses symbols to represent frequencies. It must have a **key** stating how many items one symbol represents. Parts of a symbol represent fractions of that number.

Worked Example: Key: one symbol represents 12 ice creams.

- Monday: $2\frac{3}{4}$ symbols $\Rightarrow 2.75 \times 12 = 33$ ice creams.
- To show 30 ice creams: $30 \div 12 = 2.5$, so draw 2 whole symbols and half a symbol.

Exam Tip: Choose a key that divides nicely into the frequencies. A symbol representing 4 can be split into quarters (1, 2, 3); a symbol representing 6 is hard to show as 1 or 5.

6.1.5 Pie Charts (S2)

A **pie chart** shows how a whole is split into categories. The **angle** of each sector is proportional to its frequency. The angles add up to 360° .

$$\text{Sector angle} = \frac{\text{frequency}}{\text{total frequency}} \times 360^\circ$$

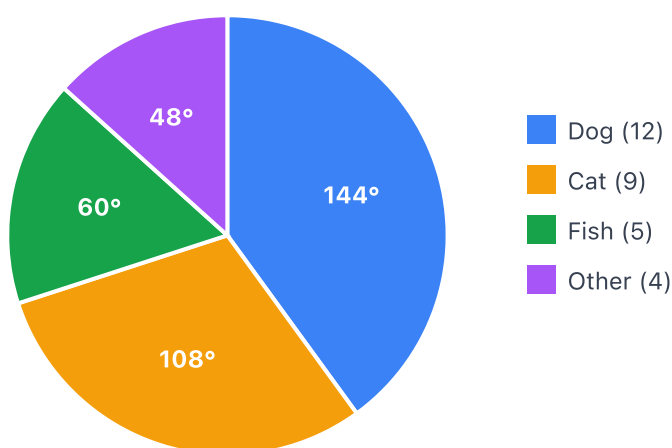
$$\text{Frequency} = \frac{\text{sector angle}}{360^\circ} \times \text{total frequency}$$

Worked Example: 30 people were asked their favourite pet: dog 12, cat 9, fish 5, other 4.

$$360 \div 30 = 12^\circ \text{ per person.}$$

Pet	Frequency	Angle
Dog	12	$12 \times 12 = 144^\circ$

Cat	9	$9 \times 12 = 108^\circ$
Fish	5	$5 \times 12 = 60^\circ$
Other	4	$4 \times 12 = 48^\circ$
Total	30	360°



Worked Example: Reading a pie chart. A pie chart represents 150 people. The sector for "bus" is 72° .

$$\frac{72}{360} \times 150 = \frac{1}{5} \times 150 = 30 \text{ people.}$$

Finding the total. A 75° sector represents 30 people. Then 1° represents $30 \div 75 = 0.4$ people, so the total is $0.4 \times 360 = 144$ people.

Worked Example: Comparing two pie charts. School P (240 students) has a 90° maths sector; School Q (600 students) has a 60° maths sector.

$$\text{P: } \frac{90}{360} \times 240 = 60 \text{ students.} \quad \text{Q: } \frac{60}{360} \times 600 = 100 \text{ students.}$$

Although P's sector is bigger, **more students in Q** chose maths. A larger angle means a larger *proportion*, not necessarily a larger number.

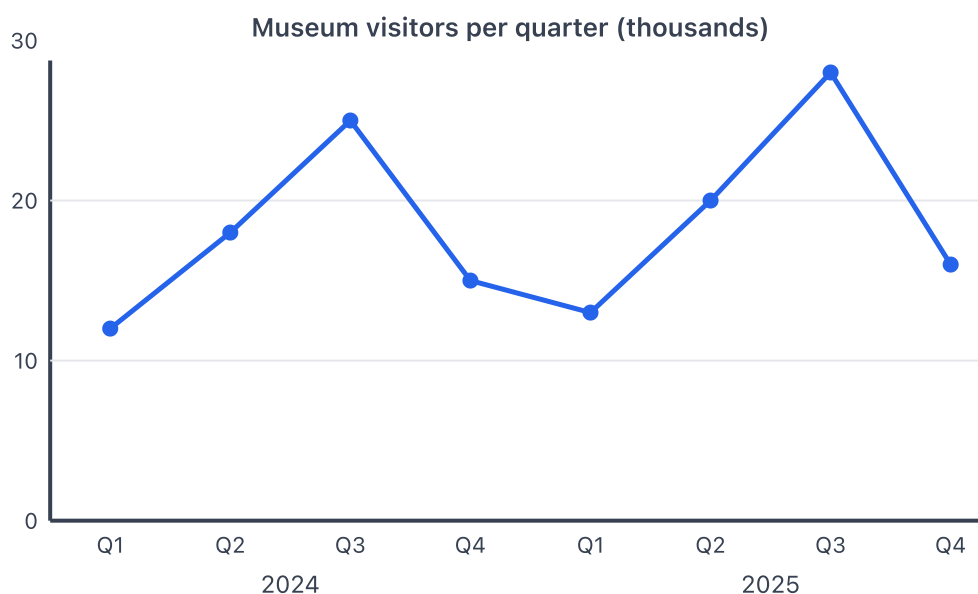
Exam Tip: Check your angles add up to 360° before drawing. Draw angles accurately with a protractor (to within $\pm 2^\circ$) and label every sector or give a key. On a non-calculator paper, find "degrees per person" ($360 \div \text{total}$) first.

6.1.6 Time Series Graphs (S2)

A **time series graph** is a line graph of data recorded at regular time intervals (days, months, quarters, years). Time goes on the horizontal axis and points are joined with straight lines.

Trend — the general direction of the data over time (rising, falling or level).

Seasonal pattern — a pattern that repeats in each cycle (e.g. highest every summer).



Worked Example: Data: 2024 — 12, 18, 25, 15; 2025 — 13, 20, 28, 16 (thousands).

Seasonal pattern: visitors peak in Q3 (summer) and are lowest in Q1 each year.

Trend: each quarter of 2025 is higher than the same quarter of 2024 (totals 70 thousand → 77 thousand), so visitor numbers are increasing.

Exam Tip: To judge a trend, compare **like with like** — the same quarter/month in different years, or yearly totals. Comparing a summer peak with a winter low tells you about the season, not the trend.

6.1.7 Choosing a Suitable Diagram (S2)

Type of data	Suitable diagrams	Example
Categorical	Bar chart, pie chart, pictogram	Favourite crisp flavour
Comparing two groups by category	Dual or composite bar chart	Boys vs girls, favourite subject
Proportions of a whole	Pie chart	How a budget is spent
Ungrouped discrete numerical	Vertical line chart, bar chart	Number of pets
Data over time	Time series line graph	Monthly rainfall
Two numerical variables (bivariate)	Scatter graph	Height and arm span
Grouped / continuous	Histogram, frequency polygon, cumulative frequency (see 6.2)	Times to run 100 m

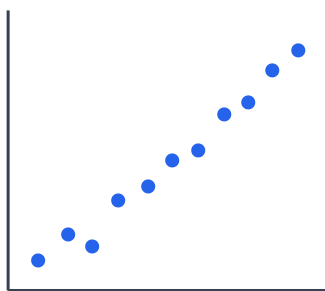
Exam Tip: Look out for **misleading** diagrams: a vertical axis that does not start at 0, uneven scales, bars of different widths, 3D effects, or pie charts with no totals. You may be asked to explain why a diagram is misleading.

6.1.8 Scatter Graphs and Correlation (S6)

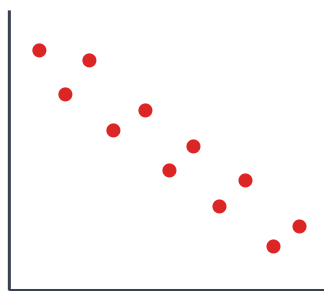
Bivariate data has two measurements for each item (e.g. hours of sunshine and number of visitors). A **scatter graph** plots each pair as a point, to look for a relationship.

Correlation describes the relationship:

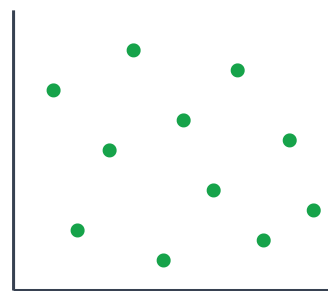
- **Positive correlation** — as one variable increases, the other tends to increase.
- **Negative correlation** — as one variable increases, the other tends to decrease.
- **No correlation** — no (linear) relationship.
- **Strong** correlation — points lie close to a straight line; **weak** — points are more spread out but still show a trend.



Strong positive



Weak negative



No correlation

Correlation does not mean causation

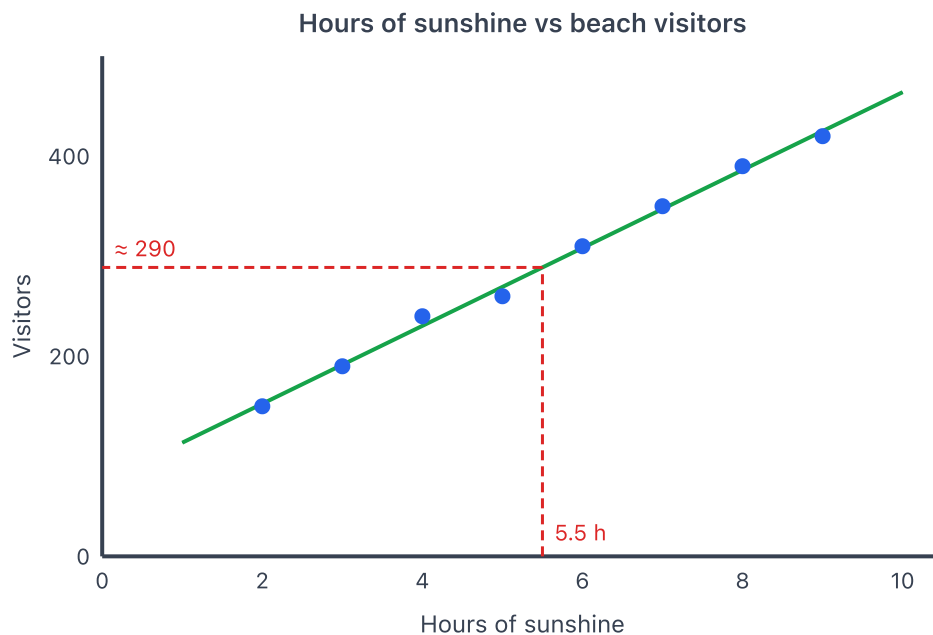
Two variables can be correlated without one causing the other. Often a **third factor** affects both.

- Ice cream sales and sunburn cases are positively correlated — both are caused by hot, sunny weather.
- Shoe size and reading ability in a primary school are positively correlated — both increase with age.

Lines of best fit

A **line of best fit** is a single straight line drawn with a ruler that follows the trend of the points, with roughly the same number of points on each side. It does not have to pass

through the origin or through any particular point. Ignore outliers when drawing it. Only draw one if there is correlation.



Worked Example: Using the line of best fit above, estimate the number of visitors on a day with 5.5 hours of sunshine.

Draw up from 5.5 on the horizontal axis to the line, then across to the vertical axis: about **290 visitors** (any value read accurately from a sensible line, roughly 270–305, is acceptable).

Interpolation and extrapolation

Interpolation — estimating a value *within* the range of the data. Usually reliable if the correlation is strong.

Extrapolation — estimating a value *outside* the range of the data. Unreliable, because the trend may not continue (e.g. a beach can only hold so many people; a day cannot have more than 24 hours of sunshine).

Worked Example: A line of best fit for fuel economy (mpg) against engine size x litres, for data from 1.0 to 3.0 litres, passes through (1.0, 58) and (3.0, 28).

$$\text{Gradient} = \frac{28 - 58}{3.0 - 1.0} = -15, \text{ so } y = -15x + c. \text{ Using } (1.0, 58): c = 73, \text{ giving } y = -15x + 73.$$

For a 2.2-litre car (interpolation): $y = -15(2.2) + 73 = 40$ mpg.

For a 5-litre car (extrapolation): $y = -75 + 73 = -2$ mpg — impossible, showing the danger of extrapolating.

The gradient means fuel economy falls by about 15 mpg for each extra litre of engine size.

Outliers

An **outlier** is a point that does not fit the pattern of the rest of the data (e.g. a student scoring 80 in maths but 30 in physics because they were ill). Identify it, suggest a reason in context, and ignore it when drawing the line of best fit.

Exam Tip: When describing correlation in an exam, give the type **and** interpret it in context: "Negative correlation — the older the car, the lower its value." When asked if a prediction is reliable, check whether it is inside the data range and how strong the correlation is.

Summary

Skill	Key point
Sampling	Random sample: everyone equally likely to be chosen. Larger random samples are more reliable.
Population estimate	$\frac{\text{number in sample}}{\text{sample size}} \times \text{population size}$

Pie chart angle	$\frac{\text{frequency}}{\text{total}} \times 360^\circ$
Pictogram	Always include a key; use fractions of a symbol.
Time series	Describe the trend and any seasonal pattern; compare like with like.
Scatter graph	Positive / negative / no correlation; strong or weak. Correlation is not causation.
Line of best fit	Interpolate (reliable) within the data range; extrapolation is risky.